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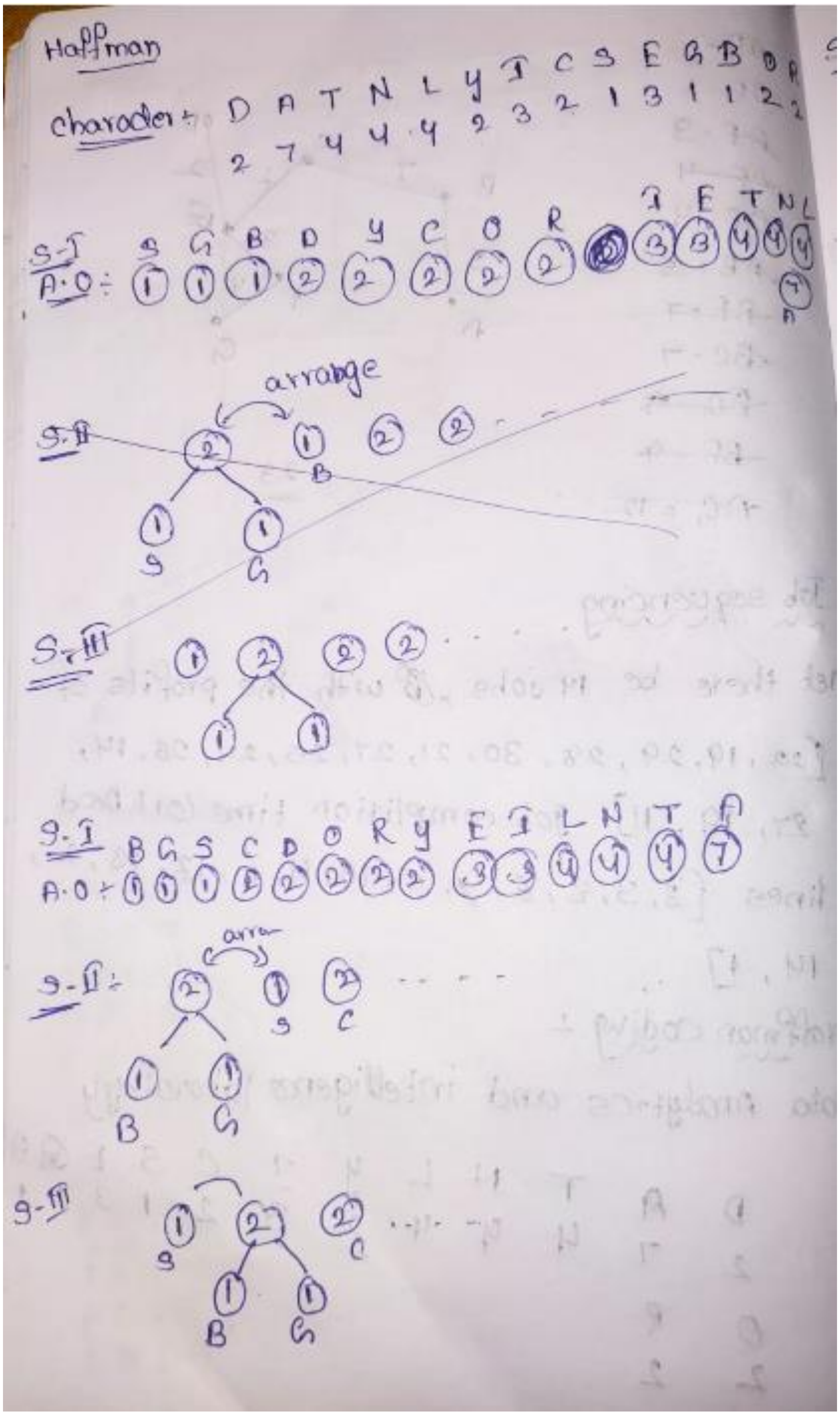
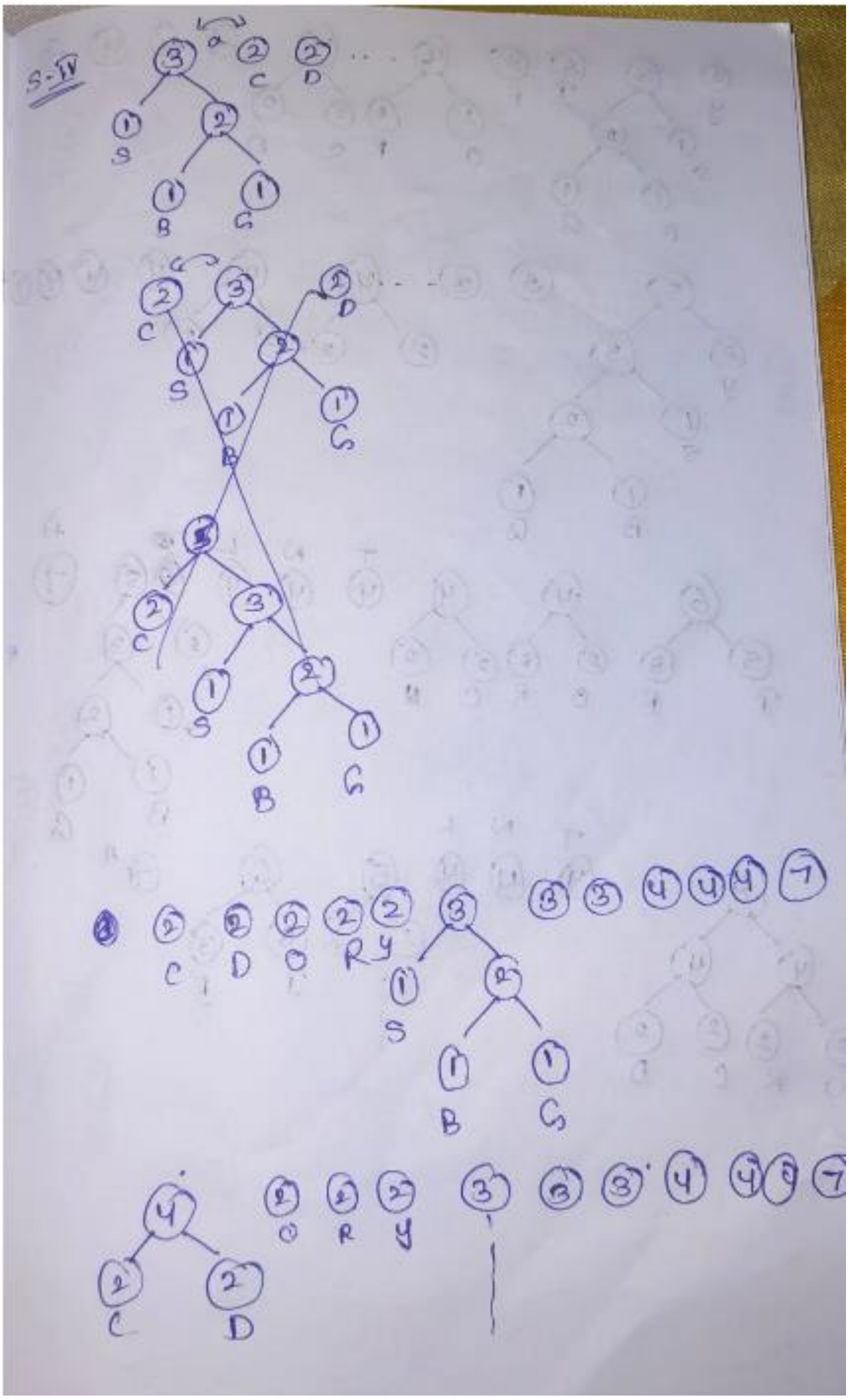
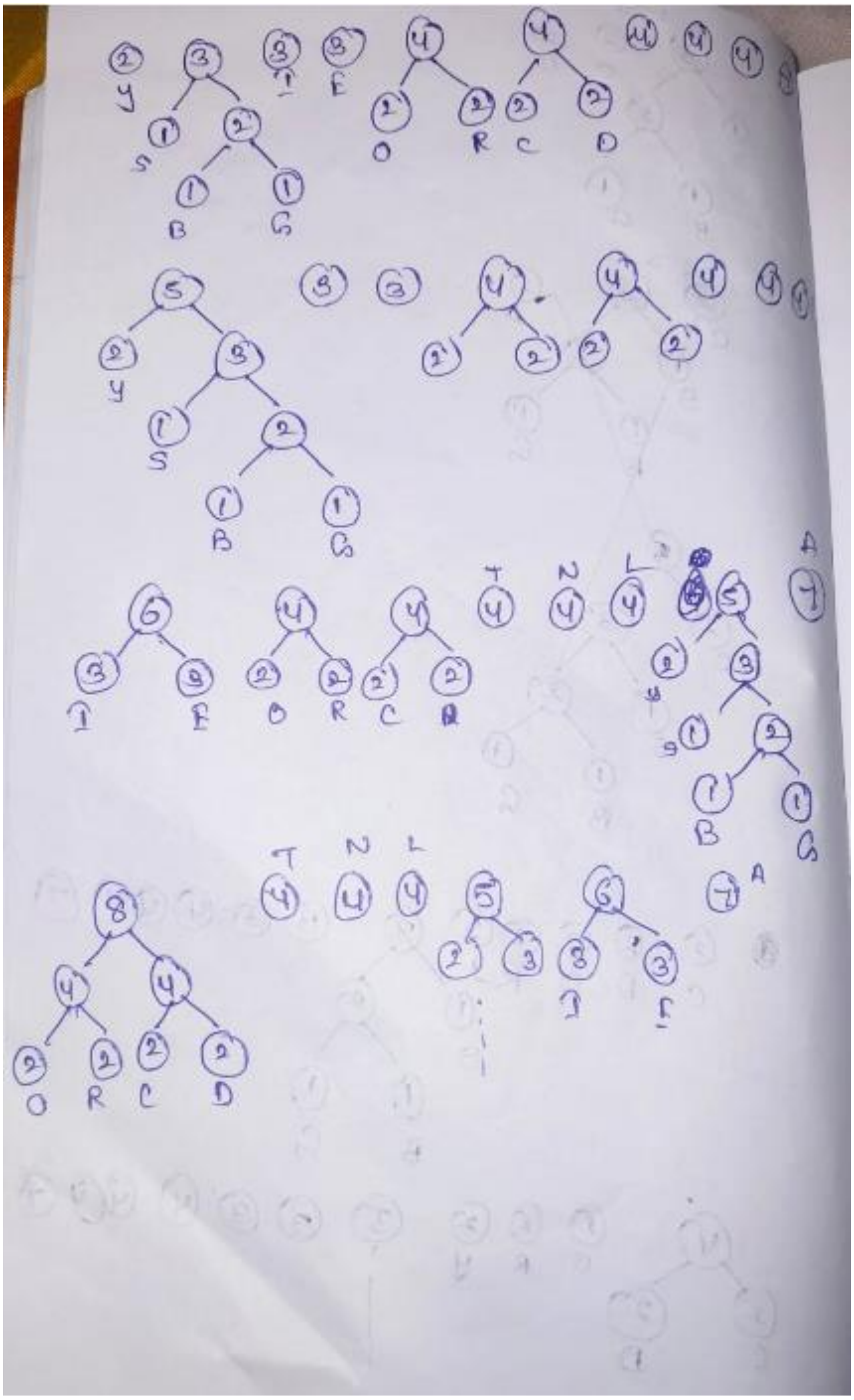
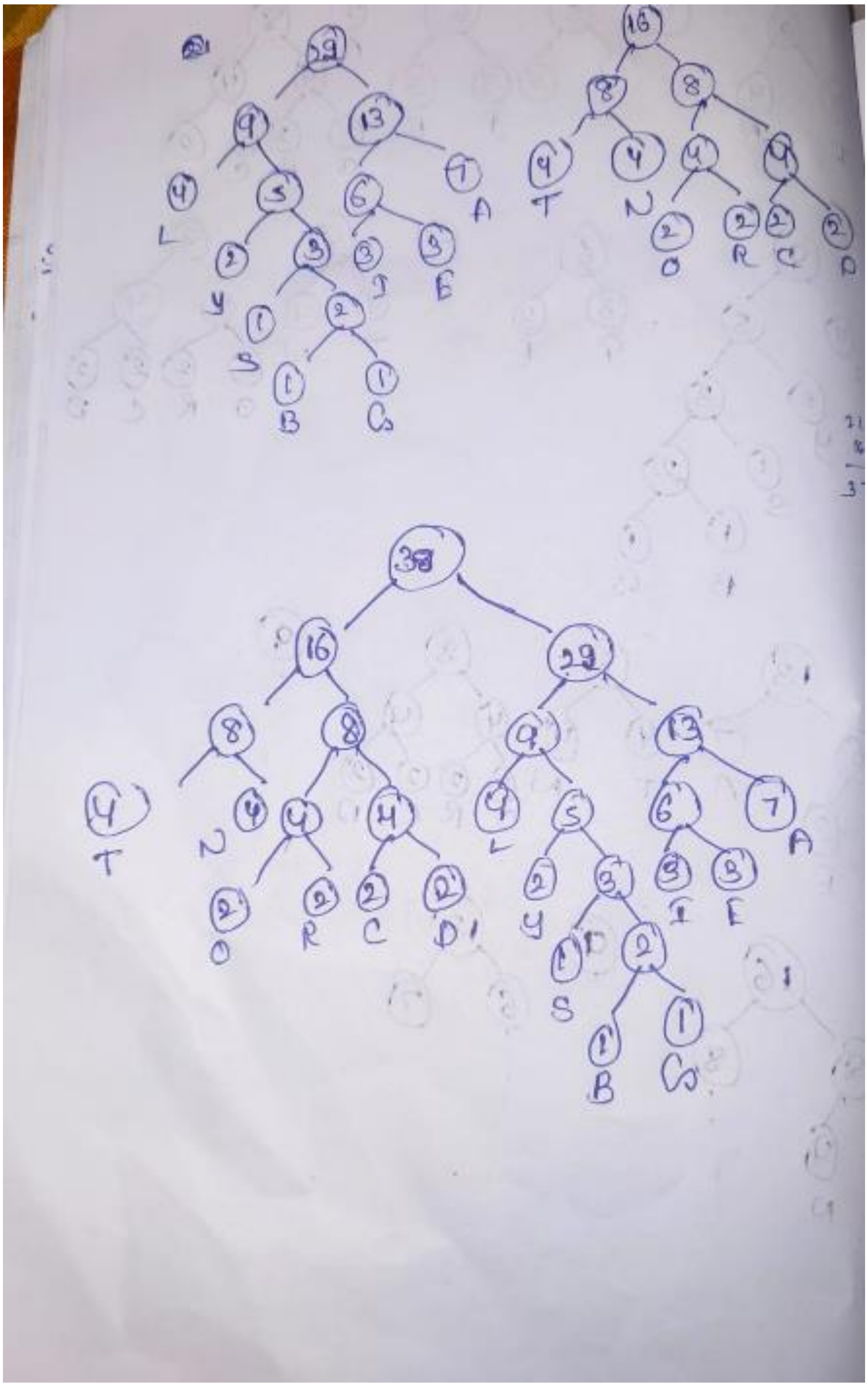
Haffman coding




S-IV





```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    char data;
    int freq;
    struct Node *left, *right;
};

struct Node* createNode(char data, int freq) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->freq = freq;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}

void swap(struct Node** a, struct Node** b) {
    struct Node* temp = *a;
    *a = *b;
    *b = temp;
}

void sortNodes(struct Node* arr[], int n) {
    int i, j;
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(arr[j]->freq > arr[j + 1]->freq) {
                swap(&arr[j], &arr[j + 1]);
            }
        }
    }
}

void printCodes(struct Node* root, int code[], int top) {
    int i;

    if(root->left != NULL) {
        code[top] = 0;
        printCodes(root->left, code, top + 1);
    }

    if(root->right != NULL) {
```

```
    int i;

    if(root->left != NULL) {
        code[top] = 0;
        printCodes(root->left, code, top + 1);
    }

    if(root->right != NULL) {
        code[top] = 1;
        printCodes(root->right, code, top + 1);
    }

    if(root->left == NULL && root->right == NULL) {
        printf("%c : ", root->data);
        for(i = 0; i < top; i++)
            printf("%d", code[i]);
        printf("\n");
    }
}

void huffman(char data[], int freq[], int size) {
    struct Node* nodes[100];
    int i;

    for(i = 0; i < size; i++) {
        nodes[i] = createNode(data[i], freq[i]);
    }

    while(size > 1) {
        sortNodes(nodes, size);

        struct Node* left = nodes[0];
        struct Node* right = nodes[1];

        struct Node* newNode = createNode('$', left->freq + right->freq);
        newNode->left = left;
        newNode->right = right;

        nodes[0] = newNode;
        nodes[1] = nodes[size - 1];
        size--;
    }

    int code[100];
```

```
    int code[100];
    printf("Huffman Codes:\n");
    printCodes(nodes[0], code, 0);
}

int main() {

    // Given Input (e frequency combined: 3 + 2 = 5)
    char data[] = {'d','a','t','n','l','y','i','c','s','e','g','b','r'};
    int freq[]  = { 2 , 7 , 4 , 4 , 4 , 2 , 3 , 2 , 1 , 5 , 1 , 1 , 2 };

    int size = 13;

    huffman(data, freq, size);

    return 0;
}
```

```
Huffman Codes:
c : 0000
r : 0001
d : 0010
y : 0011
t : 010
n : 011
l : 100
e : 101
b : 11000
s : 110010
g : 110011
i : 1101
a : 111

--------------------------------
Process exited after 0.3429 seconds with return value 0
Press any key to continue . . .
```